

Finish the test yourself before check solutions!

University of Nevada, Reno — MATH182 (Calculus 2)
Midterm 1 Sample test. Spring 2019

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Name: Solutions

To receive full credit for a problem you must show all necessary work.

1. (§5.3) Find the derivative of the function: $f(x) = \int_3^{2x^3-1} \sin(2t) dt$.

$$\text{Suppose } F'(t) = \sin 2t$$

$$\text{So } f(x) = F(2x^3-1) - F(3) \text{ by fundamental theorem.}$$

$$\begin{aligned} \text{So } f'(x) &= F'(2x^3-1) \cdot 6x - 0 \\ &= [\sin 2(2x^3-1)] \cdot 6x \end{aligned}$$

2. (§5.5) Compute the definite integral $\int_e^{e^2} \frac{1}{x(\ln x)^3} dx$

$$\text{Let } u = \ln x$$

$$du = \frac{1}{x} dx$$

$$dx = x du$$

$$u(e) = \ln e = 1$$

$$u(e^2) = \ln e^2 = 2$$

$$\int_e^{e^2} \frac{1}{x(\ln x)^3} dx$$

$$= \int_1^2 \frac{1}{x \cdot u^3} \cdot x du$$

$$= \int_1^2 u^{-3} du$$

$$= \left. \frac{u^{-2}}{-2} \right|_1^2$$

$$= \left. -\frac{1}{2u^2} \right|_1^2 = \left(-\frac{1}{8}\right) - \left(-\frac{1}{2}\right) = \frac{3}{8}$$

3. (§7.8) Find $\int_{-1}^{\infty} 6x^2 e^{-x^3} dx$.

Step 1. Let $u = -x^3$
 $du = -3x^2 dx$
 $dx = -\frac{1}{3x^2} du$

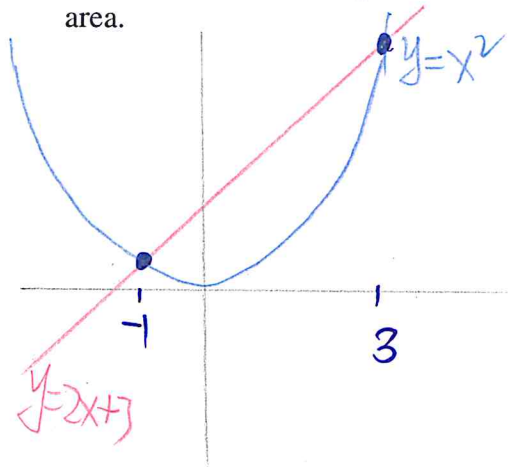
Step 3. $\int_{-1}^{\infty} 6x^2 e^{-x^3} dx$
 $= \lim_{t \rightarrow \infty} \int_{-1}^t 6x^2 e^{-x^3} dx = \lim_{t \rightarrow \infty} -2e^{-x^3} \Big|_{-1}^t$
 $= \lim_{t \rightarrow \infty} (-2e^{-t^3} + 2e) = 2e$

Step 2. $\int 6x^2 e^{-x^3} dx = \int 6x^2 e^u \cdot \left(-\frac{1}{3x^2}\right) du = -2 \int e^u du = -2e^u + C = -2e^{-x^3} + C$

4. (§11.1) Find $\lim_{n \rightarrow \infty} \frac{n}{\sqrt{3n^2 + 2}}$.

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{3n^2 + 2}} = \lim_{n \rightarrow \infty} \frac{n/n}{\sqrt{3n^2 + 2}/n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{3 + \frac{2}{n^2}}} = \frac{1}{\sqrt{3 + \lim_{n \rightarrow \infty} \frac{1}{n^2}}} = \frac{1}{\sqrt{3}}$$

5. (§6.1) Sketch the region enclosed by the parabola $y = x^2$ and the line $y = 2x + 3$, and find the area.



$$\begin{aligned} \text{Area} &= \int_{-1}^3 (2x+3) - x^2 dx \\ &= \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 \\ &= \frac{32}{3} \end{aligned}$$

intersection

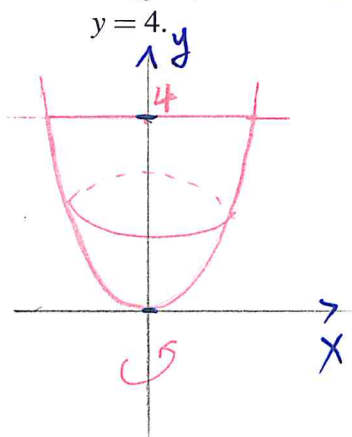
$$x^2 = 2x + 3$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3 \text{ or } -1$$

6. (§6.2) Let S be a solid obtained by rotating about the y -axis the region bounded by $y = x^4$ and



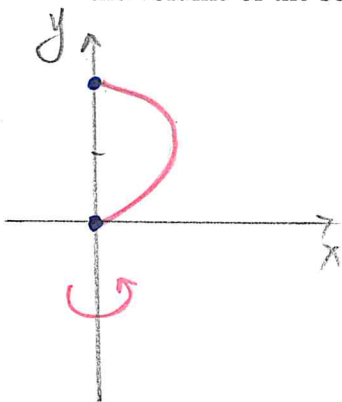
$$0 \leq y \leq 4$$

$$x = y^{\frac{1}{4}}$$

$$\begin{aligned} A(y) &= \pi x^2 \\ &= \pi \left(y^{\frac{1}{4}} \right)^2 \\ &= \pi y^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} \text{Volume} &= \int_0^4 A(y) dy = \int_0^4 \pi y^{\frac{1}{2}} dy = \frac{2}{3} \pi y^{\frac{3}{2}} \Big|_0^4 \\ &= \frac{2}{3} \pi 4^{\frac{3}{2}} = \frac{16\pi}{3} \end{aligned}$$

7. (§6.3) The region R is enclosed by the curves $x = 2y^2 - y^3$ and $x = 0$. Use cylindrical shells, find the volume of the solid obtained by rotating R about the x -axis.



intersection $2y^2 - y^3 = 0$

$$y^2(2 - y) = 0$$

$$y = 0 \text{ or } 2$$

$$\text{Volume} = \int_0^2 2\pi y (2y^2 - y^3) dy$$

$$= 2\pi \int_0^2 (2y^3 - y^4) dy$$

$$= 2\pi \left(\frac{2y^4}{4} - \frac{y^5}{5} \right) \Big|_0^2 = 2\pi \left(\frac{32}{4} - \frac{32}{5} \right)$$

$$= \frac{16\pi}{5}$$